

HARISH KUMAR G

Design Verification Engineer | UVM | SystemVerilog | Security IP | SoC

Bengaluru, India | +91 9865788557 | harishkumarg.hk98@gmail.com | linkedin.com/in/g-harishkumar | github.com/gharishkumar

PROFESSIONAL SUMMARY

RTL Design & Verification Engineer with 1.5+ years of hands-on RTL/DV experience and 5+ years of overall hardware/embedded engineering experience. Currently delivering silicon-ready RTL, SoC integration, and UVM-based verification at C-DAC Bengaluru under India's Digital India RISC-V (DIR-V) initiative. Specialized in cryptographic IP design, AXI integration, and design verification for security-critical hardware.

CORE SKILLS

- HDL & Verification: Verilog, SystemVerilog (IEEE 1800-2012), Constrained Random Verification, UVM, Cocotb, Xcelium, VCS, QuestaSim, Verilator
- Protocols: AHB, AXI, PCIe, Ethernet, I2C, SPI, UART
- Synthesis & FPGA Tools: Genus, Synopsys DC, Vivado, Quartus, Yosys
- Hardware Platforms: Basys 3 FPGA, Arty A7, VCU 118, ESP, Raspberry Pi, ATmega 328P
- Programming & Scripting: C, C++, Java, Python, Bash, TCL, Make

PROFESSIONAL EXPERIENCE

Project Associate - RTL Design & Verification - C-DAC Bengaluru

Mar 2025 - Present

Digital India RISC-V (DIR-V) Programme - BHARAT TPM

- Designed and implemented the ECDSA hardware accelerator at RTL level for NIST P-256/P-384/P-521, with block-level architecture, Verilog development, and AXI IP integration into the SoC fabric as part of India's indigenous TPM, targeting TCG TPM 2.0 compliance.
- Contributed to the Miller-Rabin primality test RTL within the RSA key generation module; applied Montgomery exponentiation at the multiplier level, reducing computational overhead by approximately 30% and improving RSA key generation throughput.
- Performed timing analysis and resource trade-off evaluation for RSA and ECC cryptographic hardware blocks.
- Analyzing the TPM 2.0 PTP PC Client Platform specification, built a complete UVM verification environment for the TPM, to drive command and capture response structures.
- Implemented constrained-random sequences with coverage-driven verification, achieving 100% of panned test vector coverage across all implemented TCG commands.
- Integrated UVM constrained-random reactive sequencer patterns for stimulus generation.
- Developed Python automation for TPM response parsing, command-response validation, and error-code decoding, improving verification turnaround by approximately 40%.
- Contributed to key provisioning flow and a post-quantum cryptography (PQC) field-upgrade flow for BHARAT TPM, ensuring forward-compatibility with emerging cryptographic standards.

R&D Engineer - Hashan Medicare

Mar 2020 - Aug 2024

- Developed firmware and selected operating systems for embedded healthcare applications.
- Scripted and automated production processes for medical devices, reducing man days by approximately 25%.
- Implemented low-power optimizations, including peripheral sleep modes and efficient memory usage, for battery-operated remote control for vision chart, extending battery life by up to 30%.
- Eliminated a separate microcontroller in Acuity Vision Chart by implementing IR decoding directly on a Raspberry Pi using GPIOs and interrupt-driven logic, reducing BOM cost by approximately 15% and product reliability increase by 60%.
- Configured hardware timers, interrupt handling, and control registers for precise timing and control for Wet Field Bi-Polar Cautry - Digital Edition.

KEY VLSI PROJECTS

- Async FIFO: Designed and verified a parameterized asynchronous FIFO supporting variable depth, using Formal verification methodology SVA, performed ASIC checks - Lint and CDC.
- Bit-Slice ALU: Developed a modular ALU (ADD, SUB, OR) with resource sharing, inspired by the IC 74181; performed timing closure and synthesis for FPGA.
- Out-of-Order AXI Verification: Verified AXI transactions using SystemVerilog, applying constrained random stimulus on address, data and ID channels.
- Floating-Point Matrix Multiplier: Designed an IEEE 754-compliant pipelined multiplier/adder; verified with Cocotb and a UVM scoreboard; synthesized and implemented on a Basys 3 FPGA.

EDUCATION

Post Graduate Diploma in VLSI Design - C-DAC ACTS, Pune

Aug 2024 - Feb 2025 | Grade: A+

Bachelor of Engineering - Electrical and Electronics Engineering, Anna University, Chennai

2016 - 2020 | CGPA: 7.4

ADDITIONAL INFORMATION

- Languages: Tamil, English, Hindi
- Interests: Calligraphy, Typography, Gardening, Strolling